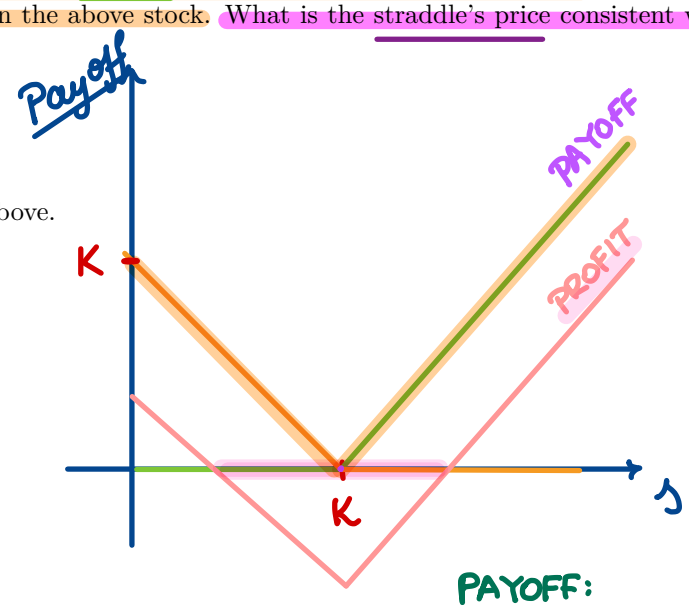
M339D: March 13th, 2026.

Problem 9.5. Consider a non-dividend paying stock whose current price is \$95 per share. You model the evolution of this stock price over the following year using a one-period binomial tree under the assumption that the stock price can be either \$120 or \$75 in one year.

The continuously compounded risk-free interest rate is $\boxed{0.06}$.

A **straddle** consists of a long call and a long otherwise identical put. Consider a $\boxed{\text{\$100-strike}}$, one-year European **straddle** on the above stock. What is the straddle's price consistent with the above stock-price model?

(a) About \$10
(b) About \$10.83
(c) About \$15.45
(d) About \$20.84
(e) None of the above.



Payoff f'tion of a straddle

$v(s) = |s-K|$

In this problem: $\boxed{K=100}$

PAYOFF:

$S(0)=95$

$S_u = 120$ $\quad V_u = 20$

$S_d = 75$ $\quad V_d = 25$

$$\Delta = \frac{V_u - V_d}{S_u - S_d} = \frac{20-25}{120-75} = -\frac{5}{45} = -\frac{1}{9}$$

$$B = e^{-rh} \cdot \frac{u \cdot V_d - d V_u}{u-d}$$

$$B = e^{-0.06} \cdot \frac{\frac{120}{\cancel{95}} \cdot 25 - \frac{75}{\cancel{95}} \cdot 20}{\frac{120}{\cancel{95}} - \frac{75}{\cancel{95}}} = e^{-0.06} \cdot \frac{120(25) - 75(20)}{45}$$

$$= 31.392$$

$$V(0) = \Delta \cdot S(0) + B = -\frac{1}{9}(95) + 31.392 = 20.84$$

Risk-Neutral Probability.

Start w/ $V(0) = \Delta \cdot S(0) + B$

$$V(0) = \frac{V_u - V_d}{\underbrace{S_u - S_d}_{\cancel{S(0)}(u-d)}} \cdot \cancel{S(0)} + e^{-rh} \cdot \frac{u \cdot V_d - d \cdot V_u}{u-d}$$

$$V(0) = \frac{1}{u-d}\left[(V_u - V_d) + e^{-rh}(u \cdot V_d - d \cdot V_u)\right]$$

$$V(0) = e^{-rh} \cdot \frac{1}{u-d}\left[e^{rh} \cdot V_u - e^{rh} \cdot V_d + u \cdot V_d - d \cdot V_u\right]$$

$$V(0) = e^{-rh} \cdot \frac{1}{u-d}\left[V_u(e^{rh} - d) + V_d(u - e^{rh})\right]$$

$$V(0) = e^{-rh}\left[\underbrace{\boxed{\frac{e^{rh}-d}{u-d}}}_{p^* :=} \cdot V_u + \underbrace{\boxed{\frac{u-e^{rh}}{u-d}}}_{=: 1-p^*} \cdot V_d\right]$$

Both positive (due to the no-arbitrage condition)!

Add up to 1!

We choose to interpret the two fractions as probabilities!
We define the risk-neutral probability of the stock price going up in a single period

$$\boxed{p^* := \frac{e^{rh}-d}{u-d}}$$

$$V(0) = \underbrace{e^{-rT}}_{\text{discounting}} \underbrace{\boxed{\left[p^* \cdot V_u + (1-p^*) \cdot V_d\right]}}_{\text{expected payoff under the risk-neutral probability}}$$

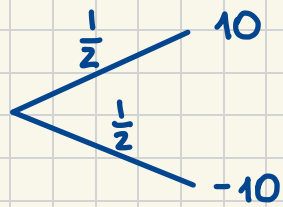
RISK-NEUTRAL PRICING FORMULA

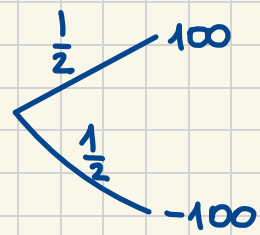
We can (and do) generalize this principle:

$$\boxed{V(0) = e^{-rh}\mathbb{E}^*[V(T)]}$$

Q: Why "risk neutral"?

Imagine bets:





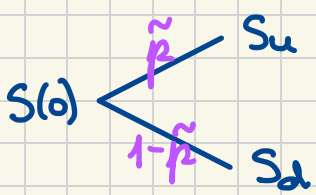
Consider a risk-neutral investor, i.e., one who is indifferent to risk and cares only about expectation.

Q: What is the probability $\tilde{p}$ such that, for a specific stock-price tree, this investor is indifferent between investing in the stock and investing in a savings account?

→: Say, we start w/ $S(0)$.

If we invest @ the ccrfir r, then our balance @ time h is $S(0)e^{rh}$.

If we invest in the stock:



$$\mathbb{E}[\text{Wealth}] = \mathbb{E}[S(h)]$$
$$= \tilde{p}\cdot S_u + (1-\tilde{p})\, S_d$$
$$= \tilde{p}\cdot u\cdot S(0) + (1-\tilde{p})\cdot d\cdot S(0)$$
$$= \left(\tilde{p}\cdot u + (1-\tilde{p})\cdot d\right)\cdot S(0)$$

$$=$$

$$\tilde{p}\cdot u + (1-\tilde{p})\cdot d = e^{rh}$$
$$\tilde{p}\,(u-d) = e^{rh} - d$$
$$\boxed{\tilde{p} = \frac{e^{rh}-d}{u-d} = p^*}$$

□

Problem 9.5 [revisited]

		Payoff
$S(0)=95$	$S_u=120$	$V_u=20$
	$S_d=75$	$V_d=25$

$r=0.06$ ✓

$$p^* = \frac{e^{rh}-d}{u-d}\cdot\frac{S(0)}{S(0)} = \frac{S(0)e^{rh}-dS(0)}{S_u-S_d} = \frac{95e^{0.06}-75}{120-75} = \underline{0.5749}$$

$$V(0) = e^{-rT}\left[V_u\cdot p^* + V_d\cdot(1-p^*)\right]$$

$$V(0) = e^{-0.06}\left[20\cdot p^* + 25\cdot(1-p^*)\right] = \underline{20.84}$$

$\square$